Chapter 1 — Differentiation: Board-Style Practice Paper with Solutions

Maharashtra HSC · Mathematics & Statistics (Arts & Science) Part II · File 3 of 3 · Chapter-wise paper (not a full official paper)

Contents

1 About this paper **1**

2 Questions **2**
2.1 Section A [16 marks] . . . 2
2.2 Section B [Attempt any six: 12 marks] . . . 3
2.3 Section C [Attempt any six: 18 marks] . . . 3
2.4 Section D [Attempt any four: 16 marks] . . . 4

3 Solutions **5**
3.1 Section A . . . 5
3.2 Section B . . . 6
3.3 Section C . . . 8
3.4 Section D . . . 10

4 Answer key **13**

5 Formula checklist (did you use each one?) **13**

6 Self-assessment guide **14**

1 About this paper

- **Format** follows the current HSC pattern (Section A: MCQ 2 marks + VSA 1 mark; Section B: 2 marks; Section C: 3 marks; Section D: 4 marks), scaled down to one chapter. A real board paper is 80 marks across all chapters; this is a **62-mark chapter-wise paper**. Suggested time: **1 hour 45 minutes**.
- **Source labels** on every question:
  - **[PYQ Mon YYYY, Qn]** — copied from the board papers in your project folder (verified against the actual papers).
  - **[Textbook]** — from the Std XII Part II textbook exercises/solved examples.
  - **[New]** — newly written in board style. Not a prediction.
- Each question shows **marks** · **difficulty** · **concept**. **Solutions start on a new page** — attempt everything first.
- No official marking schemes were in your folder. Mark splits in the solutions are **suggested** from the structure of the questions, not official.

Note: Instructions (as on the board paper)

1. Section A: Q.1 has six MCQs (2 marks each); Q.2 has four VSA questions (1 mark each). All compulsory.
2. Section B: Q.3 – Q.10, attempt **any six** (2 marks each).
3. Section C: Q.11 – Q.18, attempt **any six** (3 marks each).
4. Section D: Q.19 – Q.24, attempt **any four** (4 marks each).
5. For MCQs write the option letter **and** the answer. Only the first attempt is evaluated.

2 Questions

2.1 Section A [16 marks]

Q.1 Select and write the correct answer.

(i) If $y = \sec(\tan^{-1} x)$, then $\frac{dy}{dx}$ at $x = 1$ is _____.

(a) $\frac{1}{2}$ (b) 1 (c) $\frac{1}{\sqrt{2}}$ (d) $\sqrt{2}$

2 marks · Easy · Inverse trig + chain rule · [PYQ Feb 2026, Q1(v)]

(ii) If $x = e^{x/y}$, then $\frac{dy}{dx} =$ _____.

(a) $1 - \frac{y}{x}$ (b) $1 + \frac{y}{x}$ (c) $\frac{x-y}{x \log x}$ (d) $\frac{x+y}{x \log x}$

2 marks · Medium · Implicit/explicit conversion · [PYQ Jul 2024, Q1(v)]

(iii) Let $f(1) = 3$, $f'(1) = -\frac{1}{3}$, $g(1) = -4$ and $g'(1) = -\frac{8}{3}$. The derivative of $\sqrt{[f(x)]^2 + [g(x)]^2}$ w.r.t. x at $x = 1$ is _____.

(a) $-\frac{29}{25}$ (b) $\frac{7}{3}$ (c) $\frac{31}{15}$ (d) $\frac{29}{15}$

2 marks · Medium · Chain rule with values · [PYQ Feb 2025, Q1(v)]

(iv) If y is a function of x and $\log(x + y) = xy$, then the value of $\left(\frac{dy}{dx}\right)$ at $x = 0$ is _____.

(a) 1 (b) -1 (c) 2 (d) 0

2 marks · Medium · Implicit at a point · [PYQ Jul 2025, Q1(v)]

(v) If $y = \log_x a$, then $\frac{dy}{dx} =$ _____.

(a) $\frac{-\log a}{x(\log x)^2}$ (b) $x \log a$ (c) $-\frac{1}{\log_x a}$ (d) $\frac{1}{x \log a}$

2 marks · Easy · Change of base + chain rule · [PYQ Jun 2026, Q1(v)]

(vi) The slope of the tangent to the curve $x = \sin\theta$ and $y = \cos 2\theta$ at $\theta = \frac{\pi}{6}$ is _____.

(a) $-2\sqrt{3}$ (b) $\frac{-2}{\sqrt{3}}$ (c) -2 (d) $-\frac{1}{2}$

2 marks · Easy · Parametric + geometric meaning · [PYQ Mar 2024, Q1(v)]

Q.2 Answer the following.

(i) Find $\frac{dy}{dx}$, if $y = \sin(\log x)$. *1 mark · Easy · Chain rule · [PYQ Jul 2025, Q2(iii)]*

(ii) Find $\frac{d^3y}{dx^3}$, if $y = e^{3x+2}$. *1 mark · Easy · Higher order · [New]*

(iii) Find $\frac{dy}{dx}$, if $y = \sin^{-1} x + \cos^{-1} x$. *1 mark · Easy · Inverse trig · [New]*

(iv) Find $\frac{dy}{dx}$, if $y = \log(\sec x + \tan x)$. *1 mark · Easy · Chain rule · [New]*

2.2 Section B [Attempt any six: 12 marks]

Q.3 Find $\frac{dy}{dx}$, if $y = (\log x)^x$.
2 marks · Medium · Logarithmic differentiation · [PYQ Mar 2024, Q9]

Q.4 Find $\frac{dy}{dx}$, if $\sqrt{x} + \sqrt{y} = \sqrt{a}$.
2 marks · Easy · Implicit · [PYQ Feb 2026, Q9]

Q.5 Find $\frac{dy}{dx}$, if $x\sqrt{x} + y\sqrt{y} = a\sqrt{a}$.
2 marks · Easy · Implicit · [PYQ Jul 2025, Q9]

Q.6 Find $\frac{d^2y}{dx^2}$, if $y = x^3 + 7x^2 - 2x - 9$.
2 marks · Easy · Second order · [PYQ Feb 2026, Q10]

Q.7 If $f(x) = x^3 + x - 2$, find $(f^{-1})'(0)$.
2 marks · Medium · Derivative of inverse · [Textbook Ex. 1.2 Q4]

Q.8 Differentiate $y = \tan^{-1}\left(\frac{\cos x}{1 + \sin x}\right)$ w.r.t. x, $-\frac{\pi}{2} < x < \frac{\pi}{2}$.
2 marks · Medium · Inverse trig simplification · [Textbook Ex. 1.2 type]

Q.9 If $x = a\cos^3\theta$, $y = a\sin^3\theta$, find $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$.
2 marks · Easy · Parametric · [Textbook Ex. 1.4 Q2(ii)]

Q.10 If $e^x + e^y = e^{x+y}$, show that $\frac{dy}{dx} = -e^{y-x}$.
2 marks · Medium · Implicit · [Textbook Ex. 1.3 Q5(iv)]

2.3 Section C [Attempt any six: 18 marks]

Q.11 If $y = f(u)$ is a differentiable function of u and $u = g(x)$ is a differentiable function of x, then prove that y is a differentiable function of x and $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$.
3 marks · Medium · Proof of chain rule · [PYQ Feb 2026, Q21]

Q.12 Find $\frac{dy}{dx}$, if $y = \tan^{-1}\sqrt{\frac{3-x}{3+x}}$.
3 marks · Hard · Trig substitution · [PYQ Jul 2024, Q21]

Q.13 If $y = \log\left[\frac{x + \sqrt{x^2 + a^2}}{\sqrt{x^2 + a^2} - x}\right]$, find $\frac{dy}{dx}$.
3 marks · Hard · Rationalise + chain rule · [PYQ Jun 2026, Q21]

Q.14 Find the nth order derivative of $\log x$.
3 marks · Medium · n^{th} derivative · [PYQ Feb 2025, Q21]

Q.15 If $y = \sin^{-1} x$, then show that $(1 - x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0$.
3 marks · Medium · Second order "show that" · [PYQ Mar 2024, Q21]

Q.16 If $x^7y^5 = (x + y)^{12}$, show that $\frac{dy}{dx} = \frac{y}{x}$.

3 marks · Medium · Log + implicit · [Textbook Ex. 1.3 Q4(i)]

Q.17 Differentiate $y = x^{\sin x} + (\sin x)^x$ w.r.t. x.
3 marks · Medium · Logarithmic differentiation of a sum · [New]

Q.18 If $y = \cos(m\cos^{-1} x)$, show that $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} + m^2y = 0$.
3 marks · Hard · Second order "show that" · [Textbook Solved Ex. 1.5 (Ex. 4)]

2.4 Section D [Attempt any four: 16 marks]

Q.19 If $x = f(t)$ and $y = g(t)$ are differentiable functions of t so that y is a function of x and $\dfrac{dx}{dt} \neq 0$, then prove that $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$. Hence find $\dfrac{dy}{dx}$, if $x = at^2$, $y = 2at$.
4 marks · Medium · Parametric proof + application · [PYQ Mar 2024, Q31] (same proof: Jul 2025 Q31, Jun 2026 Q31)

Q.20 If $x = f(t)$ and $y = g(t)$ are differentiable functions of t so that y is a function of x and $\dfrac{dx}{dt} \neq 0$, then prove that $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$. Hence find the derivative of 7^x w.r.t. x^7.
4 marks · Medium · Parametric proof + one function w.r.t. another · [PYQ Feb 2025, Q31]

Q.21 If $y = f(u)$ is a differentiable function of u and $u = g(x)$ is a differentiable function of x such that the composite function $y = f[g(x)]$ is a differentiable function of x, then prove that $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$. Hence find $\dfrac{d}{dx}\left(\dfrac{1}{\sqrt{\sin x}}\right)$.
4 marks · Medium · Chain rule proof + application · [PYQ Jul 2024, Q31]

Q.22 If $x = a(\theta - \sin\theta)$ and $y = a(1-\cos\theta)$, find $\dfrac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{2}$.
4 marks · Hard · Parametric second derivative · [New]

Q.23 If $y = f(x)$ is a differentiable function of x such that $\dfrac{dy}{dx} \neq 0$ and $x = f^{-1}(y)$ exists, prove that $\dfrac{dx}{dy} = \dfrac{1}{dy/dx}$. Hence find the derivative of the inverse of $y = x^5 + 2x^3 + 3x$ at $x = 1$.
4 marks · Medium · Inverse function proof + application · [New, built on Textbook Thm 1.2.3 and Ex. 1.2 Q3(i)]

Q.24 If $y = e^{m\sin^{-1}x}$, show that $(1-x^2)\dfrac{d^2y}{dx^2} - x\dfrac{dy}{dx} - m^2y = 0$. Hence find $\dfrac{d^2y}{dx^2}$ at $x = 0$.
4 marks · Hard · Second order "show that" + evaluation · [New]

3 Solutions

3.1 Section A

Solution: Q.1 (i) — Answer: (c) $\frac{1}{\sqrt{2}}$

$$\frac{dy}{dx} = \sec(\tan^{-1} x) \cdot \tan(\tan^{-1} x) \cdot \frac{1}{1+x^2} = \sec(\tan^{-1} x) \cdot \frac{x}{1+x^2}$$

At $x = 1$: $\tan^{-1} 1 = \frac{\pi}{4}$, $\sec \frac{\pi}{4} = \sqrt{2}$.

$$\left(\frac{dy}{dx}\right)_{x=1} = \sqrt{2} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}}$$

Common Mistake: Examiner's note

Forgetting $\tan(\tan^{-1} x) = x$ or the $\frac{1}{1+x^2}$ factor. Quick check: $\sec(\tan^{-1} x) = \sqrt{1+x^2}$, whose derivative $\frac{x}{\sqrt{1+x^2}}$ gives the same value.

Solution: Q.1 (ii) — Answer: (c) $\frac{x-y}{x \log x}$

$x = e^{x/y} \Rightarrow \log x = \frac{x}{y} \Rightarrow y = \frac{x}{\log x}$.

$$\frac{dy}{dx} = \frac{\log x - x \cdot \frac{1}{x}}{(\log x)^2} = \frac{\log x - 1}{(\log x)^2}$$

Substitute $\log x = \frac{x}{y}$: $\frac{\frac{x}{y} - 1}{\frac{x^2}{y^2}} = \frac{y(x-y)}{x^2}$. Since $\frac{y}{x} = \frac{1}{\log x}$, this equals $\frac{x-y}{x \log x}$.

Common Mistake: Examiner's note

The options are in mixed x, y form, so you must convert back using $\log x = \frac{x}{y}$. Differentiating $x = e^{x/y}$ implicitly also works but is longer.

Solution: Q.1 (iii) — Answer: (d) $\frac{29}{15}$

$$\frac{d}{dx}\sqrt{f^2+g^2} = \frac{2ff' + 2gg'}{2\sqrt{f^2+g^2}} = \frac{ff' + gg'}{\sqrt{f^2+g^2}}$$

At $x = 1$: numerator $= 3\left(-\frac{1}{3}\right) + (-4)\left(-\frac{8}{3}\right) = -1 + \frac{32}{3} = \frac{29}{3}$; denominator $= \sqrt{9+16} = 5$. Value $= \frac{29}{15}$.

Common Mistake: Examiner's note

Sign slip in $(-4)(-\frac{8}{3}) = +\frac{32}{3}$ leads to option (a) or (c). These distractors are designed for that slip.

Solution: Q.1 (iv) — Answer: (d) 0

At $x = 0$: $\log y = 0 \Rightarrow y = 1$. Differentiating: $\frac{1}{x+y}\left(1 + \frac{dy}{dx}\right) = x\frac{dy}{dx} + y$. At $(0,1)$: $1 + \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = 0$.

Common Mistake: Examiner's note

Students often forget to find y at $x = 0$ first. Textbook version $\log(x + y) = 2xy$ gives 1 — read the question carefully.

Solution: Q.1 (v) — Answer: (a) $\dfrac{-\log a}{x(\log x)^2}$

$$y = \frac{\log a}{\log x} = \log a\,(\log x)^{-1} \Rightarrow \frac{dy}{dx} = -\log a\,(\log x)^{-2} \cdot \frac{1}{x} = \frac{-\log a}{x(\log x)^2}.$$

Solution: Q.1 (vi) — Answer: (c) -2

$\dfrac{dx}{d\theta} = \cos\theta$, $\dfrac{dy}{d\theta} = -2\sin 2\theta$. $\dfrac{dy}{dx} = \dfrac{-2\sin 2\theta}{\cos\theta} = \dfrac{-4\sin\theta\cos\theta}{\cos\theta} = -4\sin\theta$. At $\theta = \frac{\pi}{6}$: $-4 \cdot \frac{1}{2} = -2$.

Solution: Q.2 (i)–(iv)

(i) $\dfrac{dy}{dx} = \cos(\log x) \cdot \dfrac{1}{x} = \dfrac{\cos(\log x)}{x}$.

(ii) $y_1 = 3e^{3x+2}$, $y_2 = 9e^{3x+2}$, $\dfrac{d^3y}{dx^3} = 27e^{3x+2}$.

(iii) $\dfrac{dy}{dx} = \dfrac{1}{\sqrt{1-x^2}} - \dfrac{1}{\sqrt{1-x^2}} = 0$ (because $\sin^{-1}x + \cos^{-1}x = \frac{\pi}{2}$, a constant).

(iv) $\dfrac{dy}{dx} = \dfrac{\sec x\tan x + \sec^2 x}{\sec x + \tan x} = \dfrac{\sec x(\tan x + \sec x)}{\sec x + \tan x} = \sec x$.

3.2 Section B

Solution: Q.3 — $y = (\log x)^x$ [2 marks]

Taking log on both sides: $\log y = x\log(\log x)$. *(0.5 mark)*
Differentiating w.r.t. x:

$$\frac{1}{y}\frac{dy}{dx} = x \cdot \frac{1}{\log x} \cdot \frac{1}{x} + \log(\log x) \cdot 1 = \frac{1}{\log x} + \log(\log x) \quad \textit{(1 mark)}$$

$$\therefore \frac{dy}{dx} = (\log x)^x \left[\log(\log x) + \frac{1}{\log x}\right] \quad \textit{(0.5 mark)}$$

Common Mistake: Examiner's note

Dropping the $\frac{1}{x}$ in $\frac{d}{dx}\log(\log x)$, and leaving the answer as "$y[\ldots]$".

Solution: Q.4 — $\sqrt{x} + \sqrt{y} = \sqrt{a}$ [2 marks]

Differentiating w.r.t. x:

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0 \quad \textit{(1 mark)}$$

$$\frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}} = -\sqrt{\frac{y}{x}} \quad \textit{(1 mark)}$$

Common Mistake: Examiner's note

$\sqrt{a}$ is a constant; its derivative is 0, not $\frac{1}{2\sqrt{a}}$.

Solution: Q.5 — $x\sqrt{x}+y\sqrt{y}=a\sqrt{a}$ [2 marks]

$x^{3/2}+y^{3/2}=a^{3/2}$. Differentiating w.r.t. x:

$$\frac{3}{2}x^{1/2}+\frac{3}{2}y^{1/2}\frac{dy}{dx}=0 \quad\Rightarrow\quad \frac{dy}{dx}=-\frac{\sqrt{x}}{\sqrt{y}}=-\sqrt{\frac{x}{y}}$$

Solution: Q.6 — $y=x^3+7x^2-2x-9$ [2 marks]

$$\frac{dy}{dx}=3x^2+14x-2 \quad \textit{(1 mark)}, \qquad \frac{d^2y}{dx^2}=6x+14 \quad \textit{(1 mark)}$$

Solution: Q.7 — $(f^{-1})'(0)$ for $f(x)=x^3+x-2$ [2 marks]

Let $f(x)=0$: $x^3+x-2=0$. By inspection $x=1$ ($1+1-2=0$). (Since $f'(x)=3x^2+1>0$, f is one-one, so this is the only root.)
$f'(x)=3x^2+1 \Rightarrow f'(1)=4$.

$$(f^{-1})'(0)=\frac{1}{f'(1)}=\frac{1}{4}$$

Common Mistake: Examiner's note

Writing $\frac{1}{f'(0)}=1$ is the standard wrong answer. The 0 is a value of f, not of x.

Solution: Q.8 — $y=\tan^{-1}\left(\frac{\cos x}{1+\sin x}\right)$ [2 marks]

$$\frac{\cos x}{1+\sin x}=\frac{\cos^2\frac{x}{2}-\sin^2\frac{x}{2}}{\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)^2}=\frac{\cos\frac{x}{2}-\sin\frac{x}{2}}{\cos\frac{x}{2}+\sin\frac{x}{2}}=\frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}}=\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)$$

Since $-\frac{\pi}{2}<x<\frac{\pi}{2}$, $\frac{\pi}{4}-\frac{x}{2}\in\left(0,\frac{\pi}{2}\right)$, so $y=\frac{\pi}{4}-\frac{x}{2}$. *(1 mark)*

$$\frac{dy}{dx}=-\frac{1}{2} \quad \textit{(1 mark)}$$

Solution: Q.9 — $x=a\cos^3\theta, y=a\sin^3\theta$ [2 marks]

$\frac{dx}{d\theta}=-3a\cos^2\theta\sin\theta$, $\frac{dy}{d\theta}=3a\sin^2\theta\cos\theta$.

$$\frac{dy}{dx}=\frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta}=-\tan\theta \;\Rightarrow\; \left(\frac{dy}{dx}\right)_{\theta=\pi/3}=-\sqrt{3}$$

Solution: Q.10 — $e^x+e^y=e^{x+y}$ [2 marks]

Dividing both sides by e^{x+y}: $e^{-y}+e^{-x}=1$. Differentiating w.r.t. x:

$$-e^{-y}\frac{dy}{dx}-e^{-x}=0 \;\Rightarrow\; \frac{dy}{dx}=-\frac{e^{-x}}{e^{-y}}=-e^{y-x}$$

Exam Tip: Examiner's note

Direct differentiation also works but needs the original equation twice to simplify. The divide-first trick is cleaner and fully acceptable.

3.3 Section C

Solution: Q.11 — Proof of chain rule [3 marks]

Given: $y = f(u)$, $u = g(x)$; u is not a constant function. Let δx be a small increment in x and δu, δy the corresponding increments in u and y, with $\delta x \neq 0,\ \delta u \neq 0,\ \delta y \neq 0$.

$$\frac{\delta y}{\delta x} = \frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x}$$

Taking the limit as $\delta x \to 0$:

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \to 0} \frac{\delta u}{\delta x}$$

As $\delta x \to 0$, $\delta u \to 0$ ($\because u$ is a continuous function of x).

$$\therefore \lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta u \to 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \to 0} \frac{\delta u}{\delta x} \quad \text{.....(I)}$$

y is differentiable in u and u in x:

$$\lim_{\delta u \to 0} \frac{\delta y}{\delta u} = \frac{dy}{du}, \qquad \lim_{\delta x \to 0} \frac{\delta u}{\delta x} = \frac{du}{dx} \quad \text{.....(II)}$$

From (I) and (II): $\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{du} \cdot \frac{du}{dx}$ .....(III)
R.H.S. of (III) exists and is finite, hence the L.H.S. exists and is finite, so y is a differentiable function of x and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

Suggested split: setup & ratio (1), continuity step & limits (1), conclusion (1).

Common Mistake: Examiner's note

The most common omission is the line "$\delta u \to 0$ as $\delta x \to 0$ since u is continuous". Without it the limit change from δx to δu is unjustified.

Solution: Q.12 — $y = \tan^{-1}\sqrt{\frac{3-x}{3+x}}$ [3 marks]

Put $x = 3\cos 2\theta \Rightarrow \theta = \frac{1}{2}\cos^{-1}\left(\frac{x}{3}\right)$. For $-3 < x < 3$, $\theta \in \left(0, \frac{\pi}{2}\right)$. *(1 mark)*

$$\sqrt{\frac{3-x}{3+x}} = \sqrt{\frac{3(1-\cos 2\theta)}{3(1+\cos 2\theta)}} = \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}} = \tan\theta$$

$$y = \tan^{-1}(\tan\theta) = \theta = \frac{1}{2}\cos^{-1}\left(\frac{x}{3}\right) \quad \textit{(1 mark)}$$

$$\frac{dy}{dx} = \frac{1}{2} \cdot \frac{-1}{\sqrt{1-\frac{x^2}{9}}} \cdot \frac{1}{3} = -\frac{1}{6} \cdot \frac{3}{\sqrt{9-x^2}} = -\frac{1}{2\sqrt{9-x^2}} \quad \textit{(1 mark)}$$

Common Mistake: Examiner's note

Direct differentiation is possible but error-prone (two nested chain rules + quotient rule). If you do it directly, the final answer must still simplify to $-\frac{1}{2\sqrt{9-x^2}}$.

Solution: Q.13 — $y = \log\left[\frac{x+\sqrt{x^2+a^2}}{\sqrt{x^2+a^2}-x}\right]$ [3 marks]

Multiply numerator and denominator by $\sqrt{x^2+a^2}+x$:

$$\frac{(x+\sqrt{x^2+a^2})^2}{(x^2+a^2)-x^2} = \frac{\left(x+\sqrt{x^2+a^2}\right)^2}{a^2} \quad \textit{(1 mark)}$$

$$y = 2\log\left(x+\sqrt{x^2+a^2}\right) - 2\log a \quad \textit{(0.5 mark)}$$

Differentiating w.r.t. x ($\log a$ is constant):

$$\frac{dy}{dx} = \frac{2}{x+\sqrt{x^2+a^2}}\left(1+\frac{2x}{2\sqrt{x^2+a^2}}\right) = \frac{2}{x+\sqrt{x^2+a^2}}\cdot\frac{\sqrt{x^2+a^2}+x}{\sqrt{x^2+a^2}} = \frac{2}{\sqrt{x^2+a^2}} \quad \textit{(1.5 marks)}$$

Exam Tip: Examiner's note

Alternative: $y = \log(x+\sqrt{x^2+a^2}) - \log(\sqrt{x^2+a^2}-x)$ and differentiate each term; same answer, slightly longer. Either method gets full marks.

Solution: Q.14 — n^{th} derivative of $\log x$ [3 marks]

Let $y = \log x$.

$$y_1 = \frac{1}{x} = x^{-1}, \quad y_2 = (-1)x^{-2} = \frac{(-1)^1\,1!}{x^2}, \quad y_3 = (-1)(-2)x^{-3} = \frac{(-1)^2\,2!}{x^3}, \quad y_4 = \frac{(-1)^3\,3!}{x^4}$$

(2 marks for at least three successive derivatives written in pattern form)
In general,

$$\frac{d^n y}{dx^n} = \frac{(-1)^{n-1}(n-1)!}{x^n} \quad \textit{(1 mark)}$$

Common Mistake: Examiner's note

Writing $(-1)^n n!$ (that is the pattern for $\frac{1}{x}$). Verify with $n = 1$.

Solution: Q.15 — $y = \sin^{-1} x$ [3 marks]

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}\,\frac{dy}{dx} = 1 \quad \textit{(1 mark)}$$

Differentiating w.r.t. x:

$$\sqrt{1-x^2}\,\frac{d^2y}{dx^2} + \frac{dy}{dx}\cdot\frac{-2x}{2\sqrt{1-x^2}} = 0 \quad \textit{(1 mark)}$$

Multiplying by $\sqrt{1-x^2}$:

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = 0 \quad \textit{(1 mark)}$$

Solution: Q.16 — $x^7y^5 = (x+y)^{12}$ [3 marks]

Taking log: $7\log x + 5\log y = 12\log(x+y)$. Differentiating w.r.t. x:

$$\frac{7}{x} + \frac{5}{y}\frac{dy}{dx} = \frac{12}{x+y}\left(1+\frac{dy}{dx}\right) \quad \textit{(1 mark)}$$

$$\left(\frac{5}{y} - \frac{12}{x+y}\right)\frac{dy}{dx} = \frac{12}{x+y} - \frac{7}{x}$$

$$\frac{5x+5y-12y}{y(x+y)}\cdot\frac{dy}{dx}=\frac{12x-7x-7y}{x(x+y)}\Rightarrow\frac{5x-7y}{y}\cdot\frac{dy}{dx}=\frac{5x-7y}{x}\quad\textit{(1 mark)}$$

$$\therefore\ \frac{dy}{dx}=\frac{y}{x}\quad\textit{(1 mark)}$$

Solution: Q.17 — $y=x^{\sin x}+(\sin x)^x$ [3 marks]

Let $u=x^{\sin x}$, $v=(\sin x)^x$; $y=u+v\Rightarrow\dfrac{dy}{dx}=\dfrac{du}{dx}+\dfrac{dv}{dx}$.

$\log u=\sin x\log x\Rightarrow\dfrac{1}{u}\dfrac{du}{dx}=\dfrac{\sin x}{x}+\cos x\log x\Rightarrow\dfrac{du}{dx}=x^{\sin x}\left[\dfrac{\sin x}{x}+\cos x\log x\right]$ *(1 mark)*

$\log v=x\log\sin x\Rightarrow\dfrac{1}{v}\dfrac{dv}{dx}=x\cot x+\log\sin x\Rightarrow\dfrac{dv}{dx}=(\sin x)^x\left[x\cot x+\log\sin x\right]$ *(1 mark)*

$$\frac{dy}{dx}=x^{\sin x}\left[\frac{\sin x}{x}+\cos x\log x\right]+(\sin x)^x\left[x\cot x+\log\sin x\right]\quad\textit{(1 mark)}$$

Common Mistake: Examiner's note

Taking $\log y=\log u+\log v$ is wrong — log of a sum is not a sum of logs. Zero marks for that line onward.

Solution: Q.18 — $y=\cos(m\cos^{-1}x)$ [3 marks]

$$\frac{dy}{dx}=-\sin(m\cos^{-1}x)\cdot m\cdot\frac{-1}{\sqrt{1-x^2}}\Rightarrow\sqrt{1-x^2}\,\frac{dy}{dx}=m\sin(m\cos^{-1}x)\quad\textit{(1 mark)}$$

Squaring: $(1-x^2)\left(\dfrac{dy}{dx}\right)^2=m^2\left[1-\cos^2(m\cos^{-1}x)\right]=m^2(1-y^2)$. Differentiating w.r.t. x:

$$(1-x^2)\cdot2\frac{dy}{dx}\frac{d^2y}{dx^2}-2x\left(\frac{dy}{dx}\right)^2=-2m^2y\frac{dy}{dx}\quad\textit{(1 mark)}$$

Dividing by $2\frac{dy}{dx}$ $(\neq0)$:

$$(1-x^2)\frac{d^2y}{dx^2}-x\frac{dy}{dx}+m^2y=0\quad\textit{(1 mark)}$$

3.4 Section D

Solution: Q.19 — Parametric proof + $x=at^2, y=2at$ [4 marks]

Proof. Given $x=f(t)$, $y=g(t)$ differentiable in t and y a function of x. Let δt be a small increment in t and $\delta x,\delta y$ the corresponding increments in x,y, with $\delta t,\delta x,\delta y\neq0$.

$$\frac{\delta y}{\delta x}=\frac{\delta y/\delta t}{\delta x/\delta t}$$

Taking the limit as $\delta t\to0$; as $\delta t\to0$, $\delta x\to0$ ($\because x$ is a continuous function of t):

$$\lim_{\delta x\to0}\frac{\delta y}{\delta x}=\frac{\lim\limits_{\delta t\to0}\dfrac{\delta y}{\delta t}}{\lim\limits_{\delta t\to0}\dfrac{\delta x}{\delta t}}\quad\text{.....(I)}$$

Since x,y are differentiable in t: $\lim\limits_{\delta t\to0}\dfrac{\delta y}{\delta t}=\dfrac{dy}{dt}$, $\lim\limits_{\delta t\to0}\dfrac{\delta x}{\delta t}=\dfrac{dx}{dt}\neq0$ .....(II)

From (I), (II): $\lim_{\delta x\to 0}\dfrac{\delta y}{\delta x}=\dfrac{dy/dt}{dx/dt}$, which exists and is finite. Hence y is a differentiable function of x and $\dfrac{dy}{dx}=\dfrac{dy/dt}{dx/dt}$. *(3 marks)*

Hence. $\dfrac{dx}{dt}=2at$, $\dfrac{dy}{dt}=2a$, so $\dfrac{dy}{dx}=\dfrac{2a}{2at}=\dfrac{1}{t}$. *(1 mark)*

Other "hence" parts asked with this same proof: $y=at^2$, $x=2at\Rightarrow\frac{dy}{dx}=t$ (Jul 2025); $x=\sin t$, $y=\cos t\Rightarrow\frac{dy}{dx}=-\tan t$ (Jun 2026).

Solution: Q.20 — Parametric proof + 7^x w.r.t. x^7 [4 marks]

Proof: as in Q.19. *(3 marks)*

Hence. Let $u=7^x$, $v=x^7$. Treating x as the parameter:

$$\frac{du}{dx}=7^x\log 7,\qquad \frac{dv}{dx}=7x^6$$

$$\frac{du}{dv}=\frac{du/dx}{dv/dx}=\frac{7^x\log 7}{7x^6}=\frac{7^{x-1}\log 7}{x^6}\quad\textit{(1 mark)}$$

Solution: Q.21 — Chain rule proof + $\frac{d}{dx}\left(\frac{1}{\sqrt{\sin x}}\right)$ [4 marks]

Proof: as in Q.11. *(3 marks)*

Hence. Let $y=\dfrac{1}{\sqrt{u}}=u^{-1/2}$ where $u=\sin x$.

$$\frac{dy}{du}=-\frac{1}{2}u^{-3/2},\qquad \frac{du}{dx}=\cos x$$

$$\frac{dy}{dx}=-\frac{1}{2}(\sin x)^{-3/2}\cos x=-\frac{\cos x}{2\sin x\sqrt{\sin x}}\quad\textit{(1 mark)}$$

(Equivalent: $-\frac{1}{2}\cot x\sqrt{\operatorname{cosec} x}$.)

Solution: Q.22 — $x=a(\theta-\sin\theta), y=a(1-\cos\theta)$ [4 marks]

$$\frac{dx}{d\theta}=a(1-\cos\theta),\qquad \frac{dy}{d\theta}=a\sin\theta\quad\textit{(1 mark)}$$

$$\frac{dy}{dx}=\frac{a\sin\theta}{a(1-\cos\theta)}=\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}=\cot\frac{\theta}{2}\quad\textit{(1 mark)}$$

$$\frac{d^2y}{dx^2}=\frac{d}{d\theta}\left(\cot\frac{\theta}{2}\right)\cdot\frac{d\theta}{dx}=-\frac{1}{2}\operatorname{cosec}^2\frac{\theta}{2}\cdot\frac{1}{a(1-\cos\theta)}$$

$$=-\frac{1}{2}\operatorname{cosec}^2\frac{\theta}{2}\cdot\frac{1}{2a\sin^2\frac{\theta}{2}}=-\frac{1}{4a\sin^4\frac{\theta}{2}}\quad\textit{(1 mark)}$$

At $\theta=\frac{\pi}{2}$: $\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$, $\sin^4\frac{\pi}{4}=\frac{1}{4}$.

$$\left(\frac{d^2y}{dx^2}\right)_{\theta=\pi/2}=-\frac{1}{4a\cdot\frac{1}{4}}=-\frac{1}{a}\quad\textit{(1 mark)}$$

Common Mistake: Examiner's note

Writing $\frac{d^2y}{dx^2}=\frac{d}{d\theta}\left(\cot\frac{\theta}{2}\right)=-\frac{1}{2}\operatorname{cosec}^2\frac{\theta}{2}$ and stopping — this misses the factor $\frac{d\theta}{dx}$ and loses the last two marks.

Solution: Q.23 — Inverse function proof + application [4 marks]

Proof. Let δx be a small increment in x and δy the corresponding increment in y; $\delta x \neq 0$, $\delta y \neq 0$.

$$\frac{\delta x}{\delta y} \times \frac{\delta y}{\delta x} = 1 \Rightarrow \frac{\delta x}{\delta y} = \frac{1}{\delta y/\delta x}, \quad \frac{\delta y}{\delta x} \neq 0$$

Taking limit as $\delta x \to 0$; then $\delta y \to 0$ ($\because y$ is continuous):

$$\lim_{\delta y \to 0} \frac{\delta x}{\delta y} = \frac{1}{\lim\limits_{\delta x \to 0} \frac{\delta y}{\delta x}} \quad \text{.....(I)}$$

As y is differentiable, $\lim\limits_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{dx} \neq 0$ .....(II)

From (I), (II): $\lim\limits_{\delta y \to 0} \frac{\delta x}{\delta y}$ exists and is finite, so x is a differentiable function of y and $\frac{dx}{dy} = \frac{1}{dy/dx}$. *(3 marks)*

Hence. $y = x^5 + 2x^3 + 3x \Rightarrow \frac{dy}{dx} = 5x^4 + 6x^2 + 3$; at $x = 1$, $\frac{dy}{dx} = 14$.

$$\left(\frac{dx}{dy}\right)_{x=1} = \frac{1}{14} \quad \textit{(1 mark)}$$

Solution: Q.24 — $y = e^{m \sin^{-1} x}$ [4 marks]

$$\frac{dy}{dx} = e^{m \sin^{-1} x} \cdot \frac{m}{\sqrt{1-x^2}} = \frac{my}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}\,\frac{dy}{dx} = my \quad \textit{(1 mark)}$$

Differentiating w.r.t. x:

$$\sqrt{1-x^2}\,\frac{d^2y}{dx^2} - \frac{x}{\sqrt{1-x^2}}\frac{dy}{dx} = m\frac{dy}{dx} \quad \textit{(1 mark)}$$

Multiplying by $\sqrt{1-x^2}$ and using $\sqrt{1-x^2}\frac{dy}{dx} = my$:

$$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} = m \cdot my = m^2y \Rightarrow (1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - m^2y = 0 \quad \textit{(1 mark)}$$

Hence, at $x = 0$: $y = e^0 = 1$, $\frac{dy}{dx} = \frac{m \cdot 1}{1} = m$. Substituting:

$$(1)\frac{d^2y}{dx^2} - 0 - m^2(1) = 0 \Rightarrow \left(\frac{d^2y}{dx^2}\right)_{x=0} = m^2 \quad \textit{(1 mark)}$$

Common Mistake: Examiner's note

Students who use the quotient rule on $\frac{my}{\sqrt{1-x^2}}$ usually get lost. Clearing the root before the second differentiation is the textbook technique and is much safer.

4 Answer key

Q	Answer	Q	Answer
1(i)	(c) $\frac{1}{\sqrt{2}}$	11	Proof
1(ii)	(c) $\frac{x-y}{x\log x}$	12	$-\frac{1}{2\sqrt{9-x^2}}$
1(iii)	(d) $\frac{29}{15}$	13	$\frac{2}{\sqrt{x^2+a^2}}$
1(iv)	(d) 0	14	$\frac{(-1)^{n-1}(n-1)!}{x^n}$
1(v)	(a) $\frac{-\log a}{x(\log x)^2}$	15	Shown
1(vi)	(c) -2	16	Shown
2(i)	$\frac{\cos(\log x)}{x}$	17	$x^{\sin x}\left[\frac{\sin x}{x}+\cos x\log x\right]+(\sin x)^x[x\cot x+\log\sin x]$
2(ii)	$27e^{3x+2}$	18	Shown
2(iii)	0	19	Proof; $\frac{1}{t}$
2(iv)	$\sec x$	20	Proof; $\frac{7^x\log 7}{7x^6}$
3	$(\log x)^x\left[\log(\log x)+\frac{1}{\log x}\right]$	21	Proof; $-\frac{\cos x}{2(\sin x)^{3/2}}$
4	$-\sqrt{y/x}$	22	$-\frac{1}{a}$
5	$-\sqrt{x/y}$	23	Proof; $\frac{1}{14}$
6	$6x+14$	24	Shown; m^2
7	$\frac{1}{4}$		
8	$-\frac{1}{2}$		
9	$-\sqrt{3}$		
10	Shown		

5 Formula checklist (did you use each one?)

- ☐ Chain rule and standard composite table (Q1(i),(iii),(v); Q2; Q21)
- ☐ Inverse function rule $(f^{-1})'(k)=\frac{1}{f'(a)}$, $f(a)=k$ (Q7, Q23)
- ☐ Inverse trig derivatives + substitution $x=a\cos 2\theta$, half-angle (Q1(i), Q8, Q12)
- ☐ Logarithmic differentiation, including sums (Q3, Q16, Q17)
- ☐ Implicit differentiation, value at a point (Q1(ii),(iv); Q4, Q5, Q10)
- ☐ Parametric first and second derivatives (Q1(vi), Q9, Q19, Q22)
- ☐ One function w.r.t. another (Q20)
- ☐ Second-order "show that" by clearing roots (Q15, Q18, Q24)
- ☐ n^{th} derivative pattern (Q14)

6 Self-assessment guide

Score yourself honestly using the suggested mark splits (only count questions you attempted within the choice limits; max 62).

Score	Where you stand	What to do next
54 – 62	Board-ready for this chapter	Do Misc. Ex. 1 once under time pressure; move on
42 – 53	Solid, with leaks	List every lost mark by type (T1–T10 in File 2) and redo 3 textbook questions of each weak type
30 – 41	Methods known, execution weak	Rewrite the three proofs from memory twice; redo File 2 examples without looking
below 30	Foundations missing	Go back to File 1 sections in order; do Ex. 1.1 and 1.3 fully before retrying this paper

Warning: Non-negotiable before the exam

If you dropped marks on Q.11, Q.19, Q.20, Q.21 or Q.23, fix those first. The proof + "hence" format appeared in Section D in five of the six papers checked. Losing it costs up to 4 marks that are almost free if memorised.